HAVING COUNT (*) > 5)
GROUP BY DNUMBER;

Q. ~~Identify~~ Analyse the given relation for 1NF given phone-no is multivalued attribute.

EMPLOYEE

SSN	Ename	Phone_no	Age

Given that phone-no is a multivalued therefore the relation employee violates condition for 1NF by decomposing the relation by separating a phone number by two separate relation we get.

EMP1

SSN	EName	Age

SSN	Phone_no

Now the relations EMP1 & EMP2 satisfies the condition for 1NF. Hence the relation is in 1NF

1NF - no repeating
2NF - No partial.
3NF - No indirect.

classmate
Date 23/05
Page 2006

Q. Given that Pno is in multivalued.

Project

Pno	Pname	Plocation

Given that Pno is a multivalued therefore the relation project violates condition for 1NF.

Proj1

Pno	Pname

Proj2

Pno	Plocation

2NF

A relation schema R is in 2NF if every prime non-prime attributes A in R is fully functionally dependent on the prime key of R.

LOTS

property_id	country_name	lot#	Area	Price	Tax_rate

FD1
FD2
FD3

FD1 : {property_id} -> {county_name, lot#, Area, price, Tax_rate}
FD2 : {county_name, lot#} -> {Area, Price, Tax_rate}

FD1 does not violate 2NF as LHS consists of primary key as, property id and RHS consists of other attributes. FD2 also does not violate 2NF as LHS consists of complete candidate key country-name, lot# FD3 violates 2NF as LHS consists of partial candidate key country name

Lot2

Country-name	Tax-rate

Lot1

same

3M **Imp** Finding a Key-K of R – F

Input: A relation R & set of FD 'F' on attributes of R.

1. Set K : R
2. For each Attribute A in K
   {compute $(K-A)^+$ w.r.t F;
   if $(K-A)^+$ contains all the attributes

ID	SSN	Name	Phone	Email
--	..	--	--	--

SK = {ID} {SSN} {ID, Name} {ID, SSN} {ID, Phone}
{Name, Phone}

Q. Find the candidate keys for the given relation schema R(A, B, C, D, E, F)} with the following FD's

$A \rightarrow C$
$C \rightarrow D$
$D \rightarrow B$
$E \rightarrow F$

closure

Soln
$(A)^+ \rightarrow ACDB$
$(B)^+ \rightarrow B$
$(C)^+ \rightarrow CDB$
$(D)^+ \rightarrow DB$
$(E)^+ \rightarrow EF$ $(F)^+ \rightarrow F$
$(AB)^+ \rightarrow ABCD$ $(AD)^+ \rightarrow ABCD$
$(AF)^+ \rightarrow ABCDF$ $(AC)^+ \rightarrow ABCD$ $(AE)^+ \rightarrow ABCDEF$

From the given FD's we determine closure of all LHS attributes taking one attribute at a time if the closure result in reaching all the other attributes of the given relation we conclude it to be a key if not we next use combination of attributes. find their closure. This process has to be repeated till all the attributes of relation are reached.

∴ ~~AB is the can~~ AE is the candidate key for this.

<u>Note</u>

An attribute not present in RHS will always be a part of candidate key.

Q. R(A,B,C,D) Given FD's are

$AB \to C$ $(A)^+ \to A$

$C \to B$ $(B)^+ \to B$

$C \to D$ $(C)^+ \to BCD$

$(D)^+ \to D$

$\left.\begin{array}{l}(AB)^+ \to ABCD \\ (AC)^+ \to ABCD\end{array}\right\}$ Candidate keys

R(A,B,C,D,E,F,G,H)

$CH \to G$

$A \to BC$ $A \to ABC$

$B \to CFH$ $B \to CFH$

$E \to A$ $C \to C$

$F \to EG$ $H \to H$

$D \to D$ $E \to ABC$

$F \to EG$

For the given relational schema

Book-title	Author-name	Book-type	List price	Author affi
		publisher		

Book title → {Book type, publisher}
Book-type → List price
Author name → Author-affi

Step 1:

$A \to CDF$ $\quad (A)^+ \to ACDF$ $\quad (D)^+ \to D$
$C \to D$ $\quad (B)^+ \to BE$ $\quad (E)^+ \to E$
$B \to E$ $\quad (C)^+ \to CD$ $\quad (F)^+ \to F$

$(AB)^+ \to ABCDEF$
$(AC)^+ \to ACDF$
$(BC)^+ \to BCDE$

$\therefore (AB)^+$ is the candidate key.

Book-title	Author-name	Book-type	List price	Author-affi	Publisher

FD1
FD2
FD3

→ Since there is no multivalued attribute so 1NF is not violating.

→ 2NF. FD1 is partially dependency on key hence it violates 2NF. & similarly FD3 is also partially dependency it is also violating. 2NF

For achiv 2NF

BOOK1.

Book_title	Book_type	Publisher	List_price

FD1

FD2

BOOK2

Author_name	Author affil

FD1

BOOK3

Book title	Author name

Since book type determine list price would be broken during decomposition. We put list price along with first decomposition to maintain FD1

3NF

BOOK1

Book_type	List price

Booktitle	Booktype	Publisher

FD1

Author_name	Author Affiliation

10M Q. $R(A,B,C,D,E,F)$ & $\{A,B\}$ PK

FD's

FD1 $\{A,B\} \rightarrow C$ $(A)^+ = AD$ $(D)^+ = D$

FD2 $A \rightarrow D$ $(B)^+ = BEF$ $(E)^+ = E$

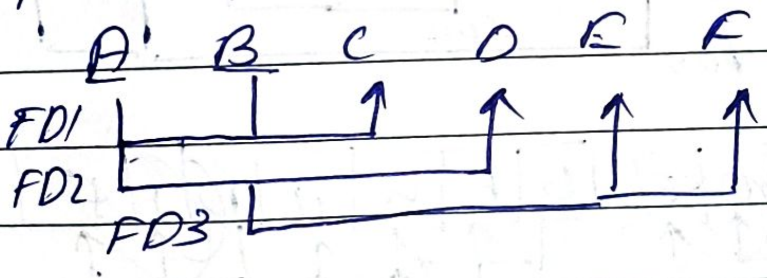
FD3 $B \rightarrow E,F$ $(C)^+ = C$ $(F)^+ = F$

$(AB)^+ = ABCDEF$

Assuming that they are single valued attribute the relation is in 1NF.

→ Analyse for 2NF

Condition for 2NF are Relation must be 1NF and there should not be partial dependancy on PK



∴ The relation can be decomposed as below.

R1

A	B	C

A	D

B	E	F

Condition for 3NF are
The relation should 2NF and no transitive dependancy.
∴ It satisfies 3NF

Q. $R(A, B, C, D, E, F, G, H, I, J, K)$

$\{A, B\} \rightarrow C$ $(A)^+ = ADEIJ$

$A \rightarrow \{D, E\}$ $(B)^+ = BFGH$

$B \rightarrow F$ $(C)^+ = C$ $(I)^+ = I$

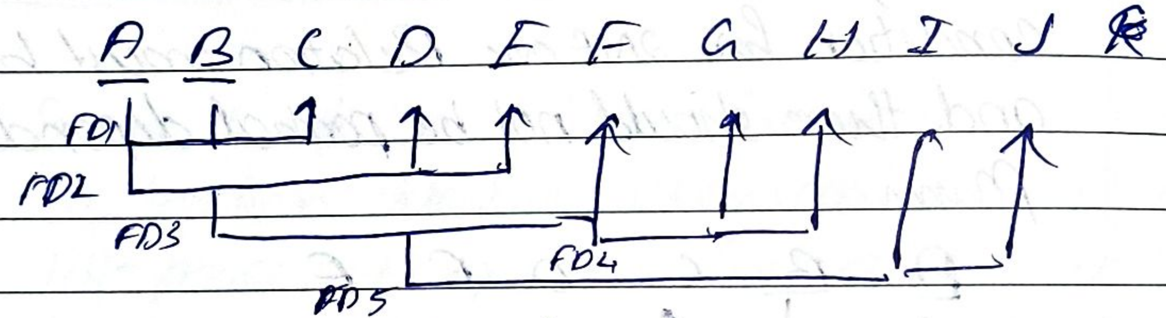
$F \rightarrow \{G, H\}$ $(D)^+ = DIJ$ $(J)^+ = J$

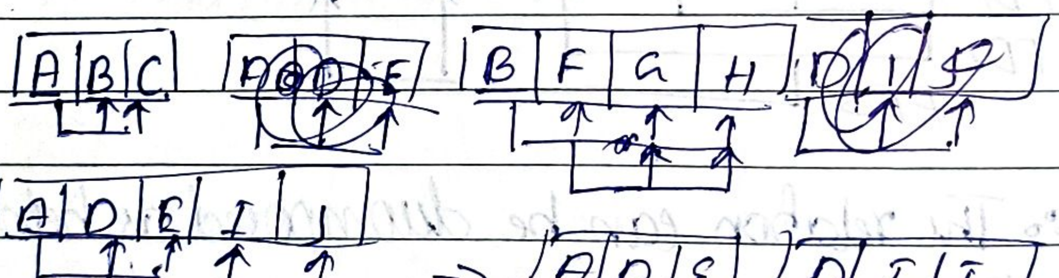
$D \rightarrow \{I, J\}$ $(E)^+ = E$ $(K)^+ = K$

$(F)^+ = GH$

Find the candidate key normalize as far as possible

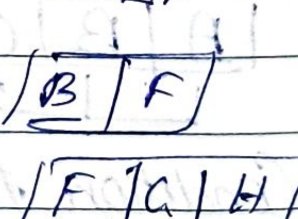
∴ AB is candidate keys.





3NF Analysis:

FD4 & FD5 violates 3NF



Normalize 3NF

REFRIG A B C D E

Model#	Year	Price	M. Plant	Colour

Model # → {M. Plant}

{Model #, year} → price

M. plant → colour

$A \to \{D\}$ $\quad$ $A \to A, D, E$

$\{A, B\} \to C$ $\quad$ $(AB)^+ \to A, B, C, D, E$

$D \to E$

A B C D E

A	D	E

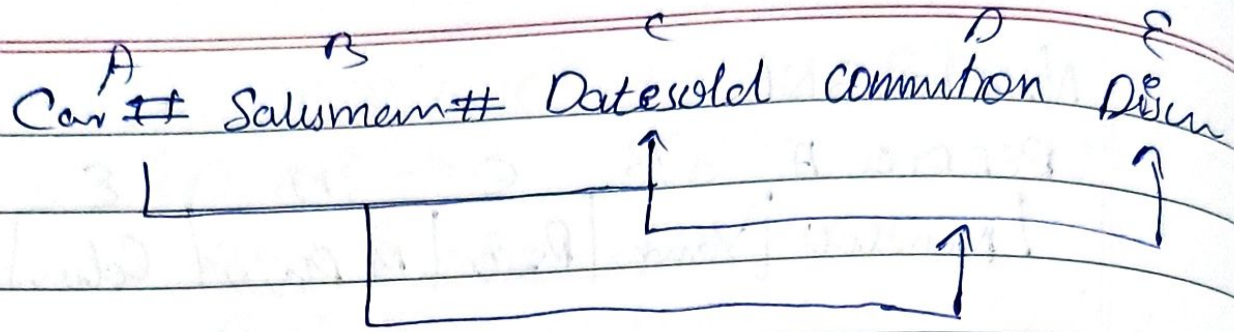
A	B	C

2NF

A	D

D	E

A	B	C

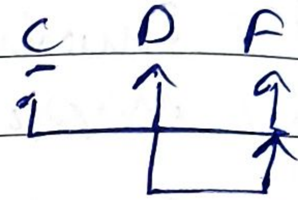


$A \to C$ $\quad A \to A, CE$

$C \to E$ $\quad B \to B, D$

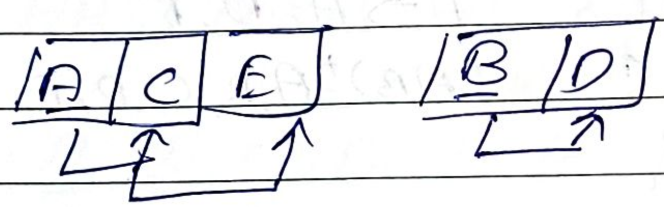
$B \to D$ $\quad C \to CE$

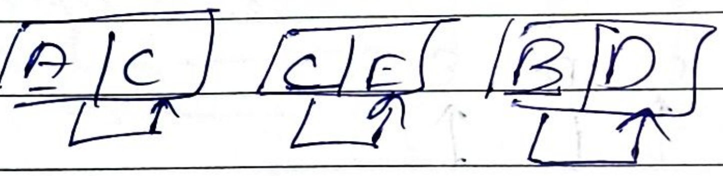
$(AB)^+ \to ABCDE$



C D F

D F





→ Minimal cover is also called Canonical cover

<u>Equalence of functional dependency</u>

Consider the two set of functional dependan-
-ces $x = \{A \to B, B \to C\}$ | $y = \{A \to B, B \to C, A \to C\}$
are they equalent.

Step 1: Checking ~~wett~~ whether all FD's of y are present in x.

LHS of y & RHS from x

a) $A \to B$

$(A)^+ = \{A B C\}$

b) $B \to C$

$(B)^+ = \{B C\}$

c) $A \to C$

$A^+ = \{A B C\}$

Step 2:

Checking whether of x are present in y.

a) $A \to B$

$A^+ = \{A B C\}$

b) $B \to C$

$B^+ = \{B, C\}$

All FD's in set x also holds in set y.

$(y \supset x)$ is true

Since all FDs in set x can be derived from y and all the FD's in set y can be derived from x.

$\therefore$ They are equivalent

$\boxed{x \equiv y}$

2) Consider the two set of FD's

$F = \{A \to C, AC \to D, E \to AD, E \to H\}$

$G = \{A \to CD, E \to AH\}$

Step 1: G in F

a) $A \to C$

$(A)^+ = \cancel{AC}\ ACD$

b) $E \to AH$

$(E)^+ = \{ACDHE\}$

Step 2:

a) $\cancel{AC} \to D \quad (A)^+ = \{ACD\}$

$F \supseteq G$ & $G \supseteq F$

$\cancel{AC}\ F = G$

3. $X = \{AB \to CD, B \to C, C \to D\}$

$Y = \{AB \to C, AB \to D, C \to D\}$

Step a) $AB \to \cancel{CD}\ C$

$(AB)^+ \Rightarrow \{ABCD\}$

b) $AB \to D$

$(AB)^+ \Rightarrow \{ABCD\}$

c) $C \to D \Rightarrow \{CD\}$

Step 2: $AB \to CD$

$AB = \{ABCD\}$ $\quad x \neq y$ 6

~~Step~~ b) ~~$AB \to A$~~

~~$AB = \{ABCD\}$~~

$B \to C$

$B = \{BCD\}$

c) $C \to D \quad C = \{CD\}$

classmate

Date
Page

4. $x = \{A \to B, B \to C, AB \to D\}$
$y = \{A \to B, B \to C, A \to C, A \to D\}$

1) in y | 2) in x.

a) $A \to B$
$(A)^+ = \{A, B, C, D\}$

b) $B \to C$
$(B)^+ = \{B\ C\}$

c) $A \to C$ d)

2) in x.
$A \to B$
$A = \{A\ B\ C\ D\}$
$B \to C$
$B = \{B\ C\}$
$AB \to D$
$(AB)^+ = \{A B C D\}$

com imp 5. <u>Minimal Cover / Canonical cover</u>

$A \to B$, $C \to B$, $D \to ABC$, $AC \to D$

Step1: Each FD must only have
Decompose RHS

$\{A \to B, C \to B, D \to A, D \to B, D \to C, AC \to D\}$

Step2: Remove the redundant key/FD's
Check if any FD can be derived from others in the set.

To find the clos $A \to B$ $A^+ = \{A\}$
not redundent key.

$C \to B$ $C^+ = \{C\}$ Not redundant
$D \to A$ $D^+ = \{D B C\}$ Not redn
$D \to B$ $D^+ = (B D A C)$ ~~Not~~ Redundant.
$D \to C$ $D^+ = \{D A B\}$ Not
$AC \to D$ $AC^+ = A\ B\ C$ Redundant. Not Redn

Step 3: Remove extraneous LHS attr

Check if A is extraneous in

$AC \to D$; find the closure of C. using current set. find the clos. of C

$(C)^+ = \{C\ B\}$

since $D \notin \{C, B\}$

If C is extraneous in $AC \to D$. find closure of A using current set

$(A^+) = \{A, B\}$

since $C \notin \{A, B\}$ hence A is not extraneous

Step 4: Minimal cover.

$\{A \to B, C \to B, D \to A, D \to C, AC \to D\}$

Q. $R = \{A, B, C, D, E, F, G, H, I, J\}$ and the set of functional dependencies:

$F = \{AB \to C, A \to DE, B \to F, F \to GH, D \to IJ\}$.

$A \to C, B \to C, A \to D, A \to E, B \to F, F \to G, F \to H, D \to I, D \to J\}$

$(AC)^+$ $A \to C$ $A^+ = \{A\ D\ E\ I\ J\}$ Not redundant

$B \to C$ $B^+ = \{B\ F\ G\ H\}$

No redundant

3) No minimum

classmate

Date ______
Page ______

Q. $R = \{A, B, C, D, E, F, G\}$

$F = \{A \to BC, BC \to F, B \to DCE, C \to DEF, E \to FG\}$

$\{A \to B, A \to C, BC \to F, B \to D, B \to C, B \to E, C \to D, C \to E, C \to F, E \to F, E \to G\}$

$A \to B$ $(A)^+ = \{A\ C\ \cancel{DEF\ FG}\}\ D\ E\ F\ G\}$

$A \to C$ ✗ $(A)^+ = \{A\ B\ \cancel{B\ C\ E\ D}\ F\ G\}$ redund

$BC \to F$ ✗ $(BC)^+ = \{B\ C\ D\ E\ F\ G$

B.

$B \to D$ ✗ $(B)^+$

$B \to C$ No

$B \to E$ ✗

$C \to D$ No

$C \to E$ No

$C \to F$ ✓

$E \to F$ No

$E \to G$ No.

$A \to B$ B

equi Q. $F = \{A \to B, B \to C, AC \to D\}$ G

$G = \{A \to B, B \to C, A \to D\}$

$A \to B$: $(A)^+ = \{A, B, C\}$ ✓

$B \to C$ ✓

$A \to D$: $\{A, B, C, D\}$

$AC \to D$ ✓

$F \equiv G$

classmate
Date
Page

Q. $F = \{A \to B, A \to C\}$
$G = \{A \to B, B \to C\}$
$A \to B \quad A^+ = \{A\ B\ C\}$ ✓
$B \to C \quad B^+ = \{B\}$ ✗
$A \to B \quad A^+ (A\ B\ C)$ ✓
$A \to C \quad A^+ (A\ B\ C)$ ✓
$\therefore F \not\equiv G$

Q.

USN	NAME	BRANCH
123	ABC	IS
456	XYZ	CS
789	MNO	IS

i) List the name & USN of IS stu
select USN, name from student where Branch = "IS"

2) List the details of stu with name XYZ
select * ~~Name~~, ~~data~~ from student where name = "XYZ"

3)

ACTOR

ACT-ID	ACT_NAME	GEN
A1	ABC	M
A2	XYZ	F
A3	MNO	M

DIRECTOR

DIR-ID	DIR-NAME	ACT-ID
D1	P	A1
D2	Q	A2
D3	R	A1
D4	S	A3

i) Retrive the names of Men

Select ACT_NAME from ACTOR WHERE GENDER='M'

ii) List the names of the actors who has been directed by more than ~~one act~~ director.

Select A.name from Actor A, Director D

Where ~~A.ACT-ID = D.ACT-ID AND ACT-ID in (Select ACT-ID from DIRECTOR having count(*) >=2)~~

count(ACT-ID) > 1 AND A.ACT-ID = D.ACT.ID

Date
Page

Q. Retrive all the name whose address contain the work Spring

Select ename ~~whose addr~~ from EMPLOYEE where Address 'LIKE % SPRING %'

Q. Find all the employees who are born during 1941
Select Ename from employee where Bdate 'LIKE ______ 1941'

Q. Show the resulting salary of every employee working on DBMS project with 10% rise.
~~update salary set EMPLOYEE set salary = salary * 1.1~~
SELECT ENAME, salary * 1.1 AS FINAL
FROM EMP E, PROJECT P, WORKS_ON W
WHERE P.NAME = 'DBMS' AND P.PNUM = W.PNUM
AND W.ESSN = E.SSN.

4. STUDENT (USN, NAME, BRANCH, PERCENTAGE)
FACULTY (FID, FNAME, DEPT, DESIGNATION, SALARY)
COURSE (CID, CNAME, FID)
ENROLL (CID, USN, GRADE)

classmate

Date
Page

i) Find the names of students who have enrolled for course '18CS53'

Select A.name from ~~student where~~ Studen A, ENROLL E where A.USN = E.USN and CID = "18CS53"

ii) Find the name of student who have enrolled for the course thaugh by ramesh

Select ~~name~~ S.NAME FROM STUDENT S, ENROLL E, FACULTY F, COURSE C where A.USN = E.USN and E.CID = C.CID and C.FID = F.FID and FNAME = "Trishul Thamni"

iii) Give a 30% hype to the salary of CSE department faculty.

update FACULTY set salary = salary * 1.3 where dept = 'CSE'

iv) List the name of the student who have enrolled for the courses of 2010 scheme

Selc S.name from STUDENT S, Enroll E

S.USN = E.USN CID LIKE '10____';

5 WORKS(Person_name, Company_name, salary)
LIVES(Person-name, street, city)
LOCATED_IN (Company-name, city)
MANAGER (Person_name, Company_name)

1. Find the names all the person who live in 'Karkala' city.
Select w.person-name from WORKS w, LIVES L where w.person-name = L.person-name and L.city = "Karkala".

2 Retrive the names of all the person of WORKS whose salary b/w 10000 & 25000

Q. SAILOR(sid, sname, rating, age)
BOAT (bid, bname, color)
RESERVES(sid, bid, date)

1) Find the name and ages of all the sailors.
Select ~~sname~~ S.sname, S.age from Sailors.

2. Find all sailors with the rating above 7
select sname from sailors where rating > 7

3. Find the name of sailor who have reserve boat no 103
Select S.sname from Sailor S, ~~Boat~~ B, Reserve R
where $\overset{S.sid}{\cancel{B.bid}}$ = R. $\overset{sid}{\cancel{bid}}$ and R.Bid = 103

4. Find the sid of the sailors who have reserve red boat
Select R.sid from Reserve R, Boat B
where R.bid = B.bid and B.color = "red".

5. Find the colors of the boat reserved by lubber

Select B.color from Boat B, Reserves R, Sailor S where B.bid = R.bid and R.sid = S.sid and S.sname = "lubber".

6. Find the name of the sailors who have reserve atleast one boat.
Select S.sname from Sailor S, Reserve R where S.sid = R.sid and cou

→ Find the nam of sailors who hav resrvd a red or a green boat.

Select S.snam from Sailor S, Reserve R, Boat B where S.sid = R.sid and R.bid = B.bid and color = "red" union color = 'green'

or

Select S.snam FROM Sailor S, Reserve R, Boat B where S.sid = R.sid and B.bid = R.bid (B.color = 'Red' or B.co = 'gr')

→ Find nam of sailor with green & red

Select S.snam from sailor S, Reserve R, Boat B where S.sid = R.sid and B.bid = R.bid and (B.color = 'Red' INTERSECT B.color = 'green')

or

(B.color = 'Red and B.color = 'green')

→ Find the sid of the all the sailor who have resrvd red but not
Select R.sid from Reserve R, Boat B where B.bid = R.bid and B.color = 'red' and B.color not in (select B.color from Boat and B.color = "green")

Find ~~sailors~~ sid of sailor who have rating of 10 or reserve boat 104

Select S.sid from sailor S, ~~Boat B~~ Reserve R where S.sid = ~~B~~ R.sid and S.rating = 10 UNION R.bid = 104.